1. Which of the following pairs correctly classifies the signal $x(t) = \cos(2\pi t) + \sin(\sqrt{2}t)$?
   - A) Periodic, Energy signal
   - B) Aperiodic, Power signal
   - C) Periodic, Power signal
   - [✓ D)] Aperiodic, Energy signal
2. Which condition must a signal $x(t)$ satisfy to be classified as an energy signal?
   - A) $0 < \int_{-\infty}^{\infty} |x(t)| dt < \infty$
   - B) $\int_{-\infty}^{\infty} |x(t)|^2 dt = \infty$
   - [✓ C)] $\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$
   - D) $x(t)$ must be periodic
3. Which of the following is a correct property of the Dirac delta function $\delta(t)$?
   - [✓ A)] $\int_{-\infty}^{\infty} \delta(t) dt = 0$
   - B) $\delta(t - t_0) = 0$ for all $t = t_0$
   - C) $\int_{-\infty}^{\infty} x(t)\delta(t - t_0) dt = x(t)$
   - D) $\int_{-\infty}^{\infty} x(t)\delta(t - t_0) dt = x(t_0)$
4. If $u(t)$ is the unit step function, then which expression represents the ramp function $r(t)$?
   - A) $\int_0^t \delta(\tau) d\tau$
   - [✓ B)] $\int_0^t u(\tau) d\tau$
   - C) $\frac{d}{dt} u(t)$
   - D) $\delta(t) * u(t)$
5. Which discrete-time signal is periodic with period N?
   - A) $x[n] = n$
   - B) $x[n] = \cos\left(\frac{2\pi n}{N}\right)$
   - C) $x[n] = e^{-n}$
   - [✓ D)] $x[n] = u[n]$
6. Which signal is obtained by sampling a continuous-time sinusoid at a frequency less than the Nyquist rate?
   - A) Oversampled
   - B) Time-scaled
   - C) Aliased
   - [✓ D)] Windowed

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7. Which of the following is a memoryless system?
   - A) $y(t) = x(t-1)$
   - B) $y(t) = x(t) + x(t-1)$
   - [✓ C)] $y(t) = x^2(t)$
   - [✓ D)] $y(t) = \int_{-\infty}^{t} x(\tau)d\tau$
8. Which of the following systems is not time-invariant?
   - [✓ A)] $y(t) = x(t-3)$
   - B) $y(t) = t \cdot x(t)$
   - C) $y(t) = x(t^2)$
   - D) $y(t) = \frac{d}{dt}x(t)$
9. A system is said to be ***BIBO stable*** if:
   - A) Every bounded input produces an unbounded output
   - B) Unbounded input produces bounded output
   - [✓ C)] Every bounded input produces a bounded output
   - D) System impulse response is discontinuous
10. Which system is linear?
   - A) $y(t) = x(t)^2$
   - [✓ B)] $y(t) = 2x(t) + 3$
   - C) $y(t) = \frac{d}{dt}x(t)$
   - D) $y(t) = \sin(x(t))$
11. Which of the following statements about LTI systems is true?
   - A) The output is always a sinusoid
   - B) The system can never be causal
   - [✓ C)] The response to any input is the same as the response to an impulse
   - [✓ D)] The response to any input can be found using convolution
12. Which of the following is *not* a property of LTI systems?
   - [✓ A)] Superposition
   - B) Time-invariance
   - C) Convolution
   - [✓ D)] Memorylessness

13. The convolution integral of two signals $x(t)$ and $h(t)$ is given by:

A) $\int_{-\infty}^{\infty} x(\tau) + h(t-\tau)d\tau$

[✓ B)] $\int_{0}^{t} x(\tau)h(\tau)d\tau$

C) $\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$

D) $\int_{-\infty}^{t} x(t)h(\tau)d\tau$

14. Let $x(t) = u(t)$ and $h(t) = e^{-t}u(t)$. What is $y(t) = x(t) * h(t)$?

A) $y(t) = u(t)$

B) $y(t) = e^{-t}u(t)$

C) $y(t) = 1 - e^{-t}$

[✓ D)] $y(t) = (1 - e^{-t})u(t)$

15. If $h(t)$ is the impulse response of an LTI system and $x(t) - \delta(t - t_0)$, then the output is:

A) $x(t - t_0)$

B) $h(t - t_0)$

C) $h(t + t_0)$

[✓ D)] $\delta(t - t_0) * h(t)$

16. An LTI system is causal if and only if:

[✓ A)] $h(t) = 0$ for $t < 0$

B) $h(t) = 1$ for $t < 0$

C) $h(t) = 0$ for $t > 0$

D) $h(t) = \delta(t)$

17. Which property is guaranteed by the commutative property of convolution?

A) $x(t) * h(t) = x(t - t_0)$

[✓ B)] $x(t) * h(t) = h(t) * x(t)$

C) $x(t) * h(t) = \delta(t)$

D) $x(t) * h(t) = x(t) + h(t)$

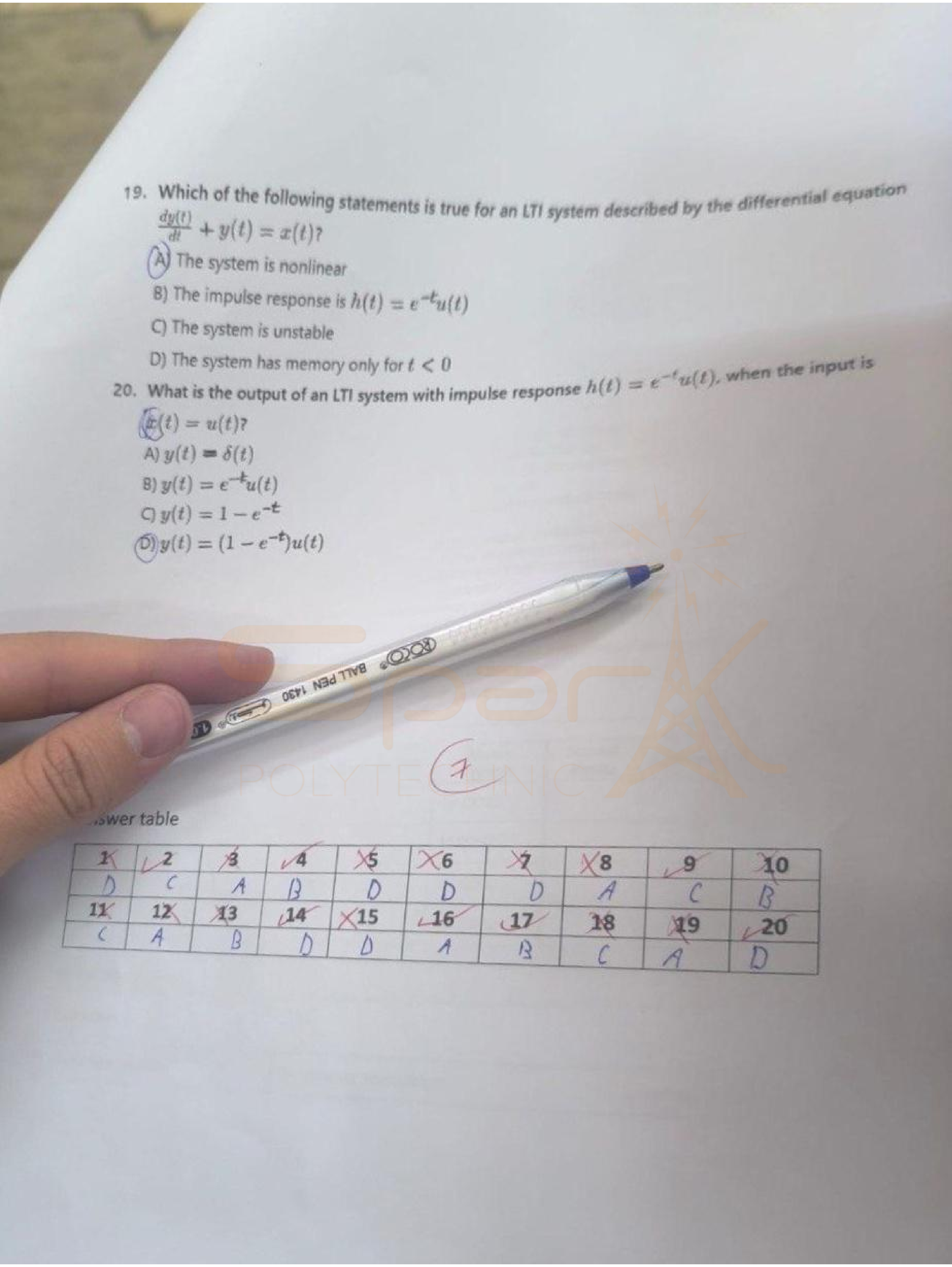
18. If an LTI system is both causal and stable, its impulse response must be:

A) Infinite for all t

B) Zero for $t < 0$ and absolutely integrable

[✓ C)] Zero for $t > 0$

D) Non-zero only at $t = 0$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19. Which of the following statements is true for an LTI system described by the differential equation $\frac{dy(t)}{dt} + y(t) = x(t)$?

[✓ A)] The system is nonlinear

B) The impulse response is $h(t) = e^{-t}u(t)$

C) The system is unstable

D) The system has memory only for $t < 0$

20. What is the output of an LTI system with impulse response $h(t) = e^{-t}u(t)$, when the input is $x(t) = u(t)$?

A) $y(t) = \delta(t)$

B) $y(t) = e^{-t}u(t)$

C) $y(t) = 1 - e^{-t}$

[✓ D)] $y(t) = (1 - e^{-t})u(t)$

7

Spark
POLYTECHNIC

swer table

1	2	3	4	5	6	7	8	9	10
D	C	A	B	D	D	D	A	C	B
11	12	13	14	15	16	17	18	19	20
C	A	B	D	D	A	B	C	A	D

Cos and sin Always power signal

1 - B
2 - C
3 - D → مش متأكد
4 - B
5 - B
6 - C
7 - C
8 - C
9 - C
10 - D
11 - D
12 - D
13 -
14 - B
15 - B

16 - A
17 - B
18 - B
19 - D
20 - D